Avijit Chowdhury

Black Hole Physics | Gravitational Waves | Astrophysical Environments

Postdoctoral Fellow, Indian Institute of Astrophysics Bengaluru, India

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Research Profile

My research focuses on black hole physics, gravitational wave phenomenology, and the influence of astrophysical environments on compact objects. I use analytical and computational methods to study black hole perturbations, quasinormal modes, accretion flows, tidal response, and the stability of compact objects in General Relativity and modified gravity. My broader goal is to understand how strong-field dynamics and realistic astrophysical environments shape gravitational wave and electromagnetic observables.

Research areas: Strong-field gravity; black hole perturbation theory; quasinormal modes and ringdown; gravitational waves; dark matter environments; accretion physics; compact objects and tidal response; modified gravity; black hole thermodynamics and Hawking radiation.

Research record: 12 journal articles, including 10 as first author; 3 communicated preprints; 1 conference-proceedings article; ANRF National Postdoctoral Fellowship; ANRF international travel support; invited lectures and presentations at national and international institutions.

Research Expertise and Contributions

- **Black hole perturbations and spectroscopy:** developed analytical and computational studies of quasinormal quasinormal modes, ringdown, stability, and echoes of compact objects. Established connections between quasinormal spectra and causal structure in modified gravity.
- **Astrophysical Environmental:** Investigated how dark matter halos modify relativistic accretion flows and black hole ringdown. Explored how ringdown and tidal response can provide complementary information about surrounding matter.
- **Modified gravity and compact objects:** investigated additional gravitational degrees of freedom, gravitational wave energy flux, and backreaction in higher derivative gravity.
- **Accretion and tidal response:** studied tidal deformation and Love numbers of accreting compact objects and connected strong-field calculations with electromagnetic and gravitational-wave signatures.

Academic Appointments

Mar 2026 – present **Postdoctoral Fellow, Indian Institute of Astrophysics**
Bengaluru, India

Apr 2024 – Feb 2026 **National Postdoctoral Fellow, Indian Institute of Technology Guwahati**
Guwahati, India

Jun 2023 – Mar 2024 **Postdoctoral Fellow, Indian Association for the Cultivation of Science**
Kolkata, India

Jun 2021 – Jun 2023 **Postdoctoral Fellow, Indian Institute of Technology Bombay**
Mumbai, India

Education

2015 – 2021 **Ph.D. in Physics, Integrated Ph.D. Program, Indian Institute of Science Education and Research Kolkata**

Advisor: Prof. Narayan Banerjee. Thesis: *On the effect of scalar fields on Hawking radiation and quasinormal modes of black holes.*

2013 – 2015 **M.Sc. in Physics, Integrated Ph.D. Program, Indian Institute of Science Education and Research Kolkata**

Advisor: Prof. Narayan Banerjee. Thesis: *On the Laws of Black Hole Mechanics for Hairy Black Holes.*

2010 – 2013 **B.Sc. (Hons.) in Physics, Asutosh College, University of Calcutta**

Kolkata, India

Fellowships, Funding and Awards

Apr 2024 – Mar 2026 **National Postdoctoral Fellowship**

Anusandhan National Research Foundation, Government of India

Jun 2025 **International Travel Support for GR24 and Amaldi16**

Anusandhan National Research Foundation, Department of Science and Technology, Government of India

Aug 2021 **FORDECYT–PRONACES–CONACYT Postdoctoral Grant**

Institute for Nuclear Sciences, National Autonomous University of Mexico; awarded but not availed

Publications

Journal Articles

12. A. Bhowmik, **A. Chowdhury** and S. Chakrabarti, "The Ringdown and the Tide: Fingerprints of Dark Matter Halo Profiles," *Physical Review D* **114**, 084070 (2026).

11. **A. Chowdhury**, G. Sen, S. Chakrabarti and S. Das, "Effect of generic dark matter halo on transonic accretion onto galactic black holes," *Physical Review D* **112**, 064041 (2025).

10. R. Ghosh, A. K. Mishra and **A. Chowdhury**, "Birkhoff's Theorem and Uniqueness: A peek beyond General Relativity," *Physics Letters B* **871**, 139974 (2025).

9. **A. Chowdhury**, A. K. Mishra and S. Chakraborty, "Connecting quasi-normal modes with causality in Lovelock theories of gravity," *Physics Letters B* **861**, 139219 (2025).

8. **A. Chowdhury**, S. Biswas and S. Chakraborty, "Accreting Schwarzschild-like compact object: Plasma-photon interaction and stability," *Physical Review D* **110**, 064072 (2023).

7. **A. Chowdhury**, S. Xavier and S. Shankaranarayanan, "The dominating mode of two competing massive modes of quadratic gravity," *Scientific Reports* **13**, 8547 (2023).

6. **A. Chowdhury**, S. Devi S. Chakrabarti, "Naked singularity in 4D Einstein-Gauss-Bonnet novel gravity: Echoes and instability," *Physical Review D* **106**, 024023 (2022).

5. **A. Chowdhury** and N. Banerjee, "Echoes from a singularity," *Physical Review D* **102**, 124051 (2020).

4. **A. Chowdhury** and N. Banerjee, "Greybody factor and sparsity of Hawking radiation from a charged spherical black hole with scalar hair," *Physics Letters B* **805**, 135417 (2020).

3. **A. Chowdhury**, "Hawking emission of charged particles from an electrically charged spherical black hole with scalar hair," *European Physical Journal C* **79**, 928 (2019).

2. **A. Chowdhury** and N. Banerjee, "Superradiant stability of mutated Reissner–Nordström black holes," *General Relativity and Gravitation* **51**, 99 (2019).

1. **A. Chowdhury** and N. Banerjee, "Quasinormal modes of a charged spherical black hole with scalar hair for scalar and Dirac perturbations," *European Physical Journal C* **78**, 594 (2018).

Preprints and Manuscripts Under Review

3. S. Sarkar†, S. Gera†, **A. Chowdhury**†, C. Singha†, S. M. Dasan†, and S. Chakraborty, "Quasinormal modes of Kerr–Newman–de Sitter black hole: Linear stability and strong cosmic censorship conjecture", arXiv:2611.09606 [gr-qc].†These authors contributed equally and share first authorship.

2. R. Ghosh, N. Speeney, **A. Chowdhury**, C. Singha, M. A. S. Pinto and E. Berti, "From Density to Mass: A New Parametrized Framework for Dark Matter Environment Around Black Holes", arXiv:2609.35771 [gr-qc].

1. **A. Chowdhury**, C. Singha, K. Bamba and S. Chakraborty, "Tidal deformation of an accreting compact object", arXiv:2607.24938 [gr-qc].

Conference Proceedings

- "Dark matter halos and transonic accretion flow," accepted for publication in *Journal of Physics: Conference Series*, proceedings of GR24 and Amaldi16, J. Phys. Conf. Ser. **3177** (2026), 012038.

Teaching Experience

Independent Instruction and Advanced Schools

Oct 2026 **Instructor, "Black Hole Perturbation Theory," IUCAA Pune**

Delivered lectures at "Advanced Topics in General Relativity for Multimessenger Observations," for Ph.D. students, postdoctoral researchers, and faculty members.

Mar 2025 **Teaching Assistant, "Beyond the Horizon: Testing the Black Hole Paradigm," ICTS Bengaluru**

Instructor: Prof. Dimitrios Psaltis; advanced school for graduate students and postdoctoral researchers.

Oct 2023 **Teaching Assistant, "Black Holes in General Relativity and beyond," ICTS Bengaluru**

Instructor: Prof. Thomas Sotiriou; 2nd IAGRG School.

Teaching Assistantships

- **IIT Bombay:** Teaching Assistant for *UG First Year Physics Lab* (PH117), Nov 2022–Feb 2023 and Apr–Jun 2023; Teaching Assistant for *Basics of Electricity and Magnetism* (PH108), Mar–Jul 2022.
- **IISER Kolkata:** Teaching Assistant for *Basic Statistical Mechanics* (PH3201), Jan–May 2019; *Basics of Field Theory and Relativistic Quantum Mechanics* (PH4106), Aug–Dec 2018; *Physics IV* (PH2201), Jan–May 2018; *Intermediate Electricity and Magnetism* (PH3202), Jan–May 2016; *Intermediate Quantum Mechanics* (PH3102), Aug–Dec 2015.

Student Mentoring

2022 – 2024 **Saraswati Devi, doctoral candidate, IIT Guwahati**

Current position: Assistant Professor, Department of Physics, Mangaldai College, Tezpur, Assam.

2023 – 2024 **Surajit Ghosh, Master's candidate, IACS Kolkata**

Current position: Graduate student, Wake Forest University, USA.

Selected Invited Presentations and Lectures

- **Jun 2026:** "Probing Dark Matter Around Black Holes: Multi-Messenger Signatures from Black Hole Accretion and Ringdown Spectroscopy," Symphony of Spacetime, Raman Research Institute, Bengaluru.
- **Oct 2025:** "Dark matter halo and transonic accretion flow," 10 Years of Gravitational Wave Discovery: Advancing the Frontiers of Gravitational Physics, IIT Gandhinagar.
- **Jul 2025:** "Effect of generic dark matter halo on transonic accretion onto galactic black holes," School of Physics and Astronomy, University of St Andrews, Scotland, UK.
- **Jul 2024:** "EFT approach to QNMs in Modified Gravity," Testing Gravity with Multimessenger Astronomy, IIT Bombay.
- **Feb 2024:** Lecture, "Black Hole Perturbations, Quasinormal Modes and Quasibound States," NIT Rourkela.
- **May 2022:** "Perturbation of spacetimes with naked singularity," Instituto de Ciencias Nucleares, UNAM, Mexico City.
- **Dec 2020:** "Scalar fields, Quasinormal modes and echoes," Centre for Theoretical Studies, IIT Kharagpur.

Selected Contributed Presentations

- **May 2026:** Poster, "Effect of generic dark matter halo on transonic accretion onto galactic black holes," 44th ASI meeting, IIT Guwahati.
- **Jul 2025:** Oral, "Effect of generic dark matter halo on transonic accretion onto galactic black holes," GR24 and

Amaldi16, Glasgow, UK.

- **Jan 2025:** Oral, "Accreting Schwarzschild-like compact object: Plasma-photon interaction and stability," 33rd IAGRG meeting, BITS Pilani.
- **Mar 2023:** Oral, "How a non-detectable mode can provide a hint of modified gravity?," 41st ASI meeting, IIT Indore.
- **Dec 2022:** Oral and poster, "Massive spin-2 mode carries more energy than spin-0 mode in quadratic gravity," 32nd IAGRG meeting, IISER Kolkata.
- **Jul 2022:** Oral, "Massive tensor modes carry more energy than scalar modes in quadratic gravity," GR23, Chinese Academy of Sciences.

Academic Service and Outreach

Journal referee: Reviewer for international scientific journals since 2022, including *Physical Review Letters*, *Physical Review D*, *European Physical Journal C*, and *General Relativity and Gravitation*.
Professional membership: Life Member, Indian Association for General Relativity and Gravitation (IAGRG).
Outreach: Weekend lecture, "Black holes as cosmic laboratories for fundamental physics," Scientific and Critical Research Institute, Kolkata, Jan 2026.

Academic and Event Organization

- Member, organizing committee for "Current Trends in Modern Physics," Department of Physical Sciences, IISER Kolkata, 2016–2017.
- Member, local organizing committee for the 4th Topical Conference on Gravity, Cosmology, Astronomy and Astrophysics, IISER Kolkata, 2015.

Computational Skills

Programming: Python (NumPy, SymPy, Bilby, Dynesty), Fortran, C, Java.
Scientific software: Mathematica (xAct), MATLAB.

Complete list of referees

- Narayan Banerjee
  Professor (Retd.),
  Department of Physical Sciences,
  Indian Institute of Science Education and Research Kolkata,
  Mohanpur, West Bengal, India
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  ✉ banerjee.narayan@gmail.com

- Sayan Chakrabarti
  Associate Professor,
  Department of Physics,
  Indian Institute of Technology Guwahati,
  Guwahati, Assam, India
  ✉ sayan.chakrabarti@iitg.ac.in
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- Sumanta Chakraborty
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  School of Physical Sciences,
  Indian Association for the Cultivation of Science,
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  Department of Physics,
  Indian Institute of Technology Bombay,
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  Department of Physics,
  Indian Institute of Technology Guwahati,
  Guwahati, Assam, India
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- Kazuharu Bamba
  Professor
  Faculty of Symbiotic Systems Science,
  Fukushima University,
  Fukushima 960-1296, Japan
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